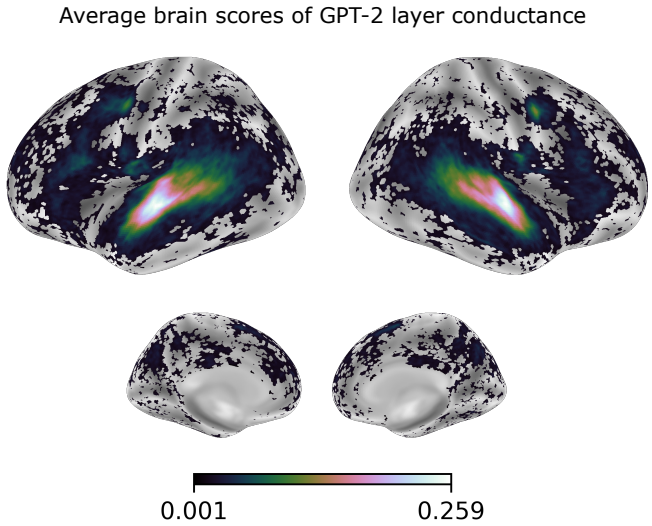

Fig. S5 Brain scores predicted by GPT-2 layer conductance. Brain scores were computed independently for the conductance of each GPT-2 layer, for each voxel, and for each individual. The statistical test (Wilcoxon signed-rank) was applied separately for each layer, and voxels significantly predicted by at least one layer were color-coded (FDR corrected, $P < 0.05$). The color bar represents the average brain score across all layers.

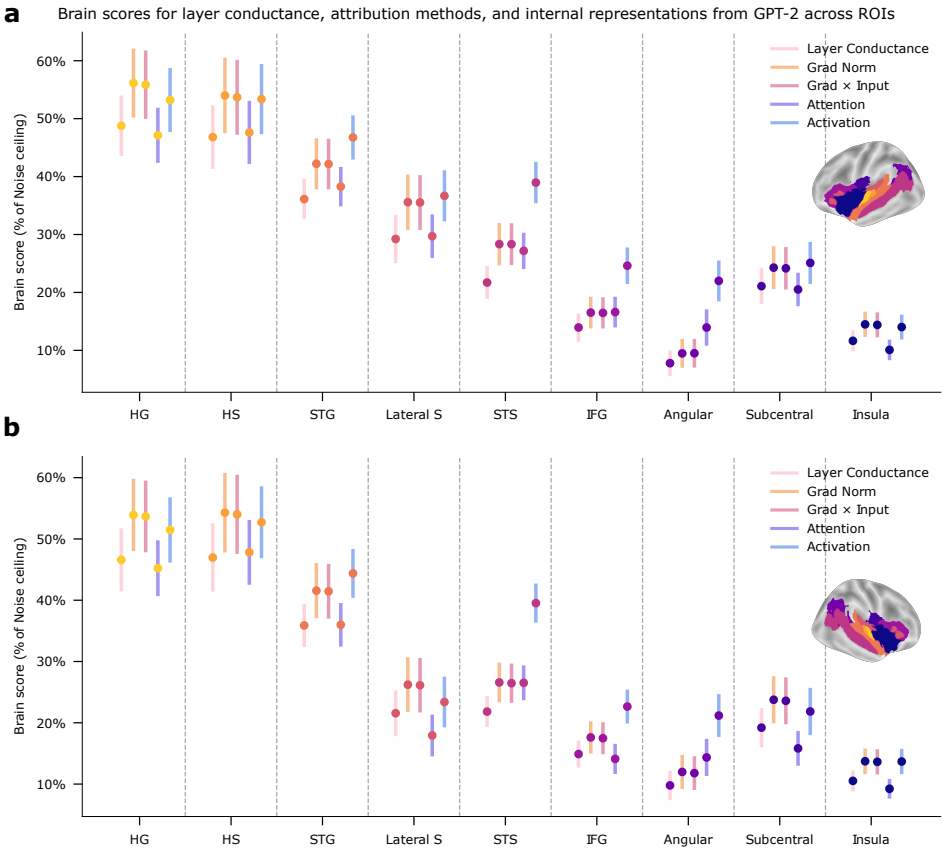

Fig. S6 Comparing brain scores of layer conductance with attribution methods and internal representations for GPT-2 across ROIs, presented separately for the left and right hemispheres. a, Brain scores for the left hemisphere. **b,** Brain scores for the right hemisphere. Brain scores were normalized and presented as a percentage of the noise ceiling. Brain scores were calculated independently for each combination of feature space and individual, then averaged across individuals. For layer conductance, scores were averaged across all layers of GPT-2. Markers represent the mean brain score (as a percentage of the noise ceiling) for each ROI. Error bars indicate the 95% confidence interval across individuals. Marker colors correspond to brain regions, while error bar colors represent different feature spaces.

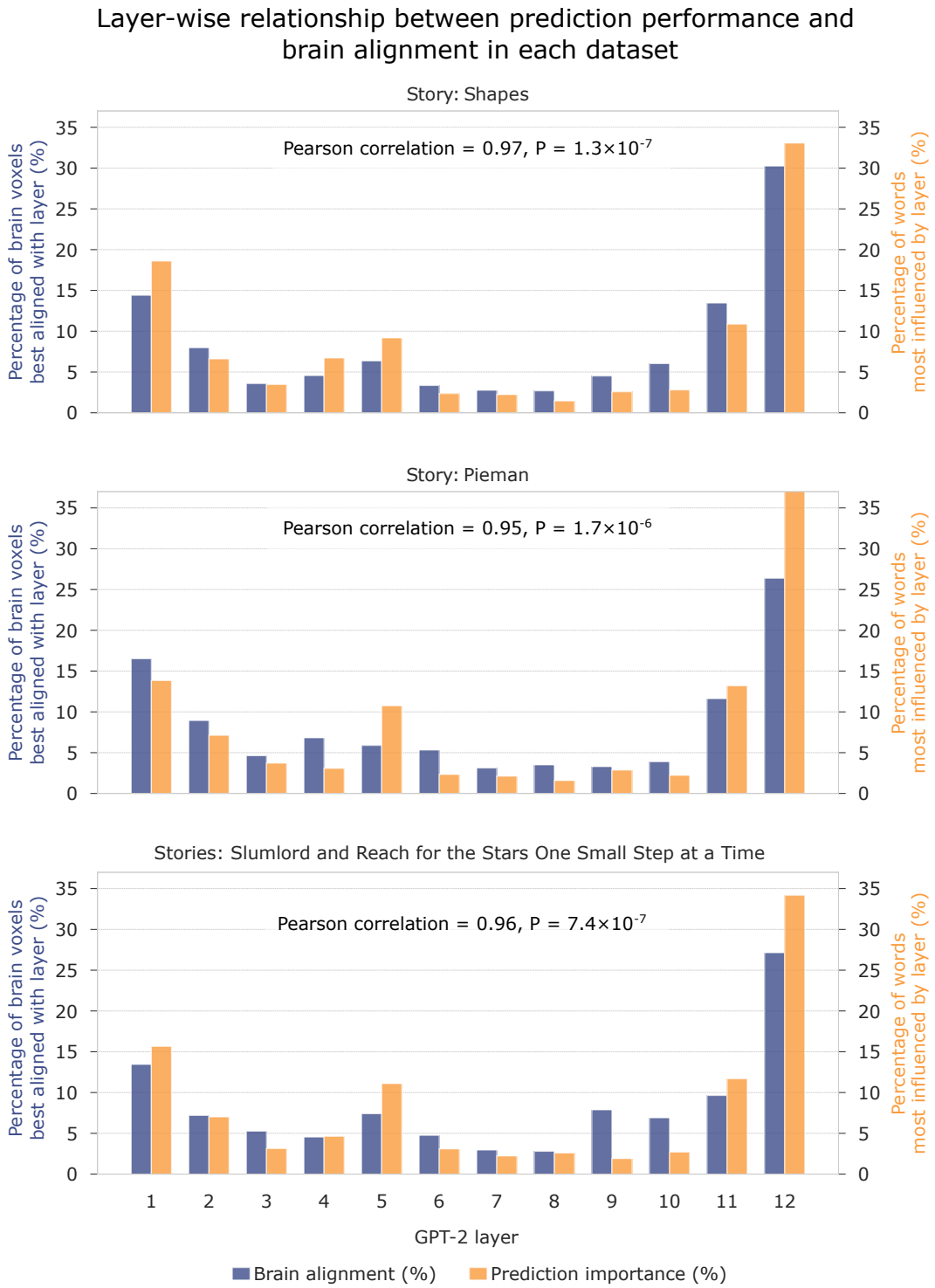

Fig. S7 Relationship between layer-wise prediction performance and brain alignment across datasets. The percentage of brain voxels aligned with each layer (blue) was compared with the percentage of words most influenced by each layer (orange), based on conductance scores. Results are shown separately for each dataset, where each dataset corresponds to a story presented both to the language model (for next-word prediction) and to human participants (during fMRI recordings). The correlation within each dataset reflects the relationship between the importance of model layers in processing the story for next-word prediction and their predictive performance in modeling brain activity for the same story.